FEASIBILITY STUDY · HARDWARE-CENTERED PAUSE

Hardwired pause

This study asks whether a long pause in new frontier training could be made technically verifiable. It analyzes a hypothetical agreement rather than endorsing the pause as policy.

Background and provenance

WHY IT WAS WRITTEN

The group was formed to test whether a mutually verifiable, long-duration pause in frontier training could actually work. Its opening scenario imagines political leaders asking advisers for a feasible design. The paper distinguishes that technical question from whether a pause is desirable or politically likely. It then studies one hardware-centred design in enough detail to estimate chip inventories, replacement timelines, and possible evasion capacity.

PEOPLE AND ORGANISATIONS

The Working Group on AI Pause Feasibility was organised by UC Berkeley professors William Fithian (statistics) and Wesley H. Holliday (philosophy). The listed contributors span UC Berkeley, Princeton, Stanford, Tufts, MIT, Mila/Université de Montréal, the Oxford China Policy Lab, and other research settings. This is a multi-institution group of individuals, not a single university policy. The paper explicitly says affiliation does not imply an institution's endorsement or that every co-author agrees with every claim.

KEY DATES

The group began weekly political and technical meetings in May 2026 and held a Berkeley workshop with panel discussions and tabletop exercises on 7 July. The initial public working paper appeared on 9 October 2026; version 1.0.1 followed on 10 October with reproducibility material, and version 1.0.2 on 26 October with corrections and updated references. This brief uses version 1.0.2.

EARLIER FOUNDATIONS

The introduction traces pause advocacy back to the March 2023 open letter, while the design cites earlier work on inference-only and model-restricted chips. Those are antecedents, not sections of a 2023 Hardwired Pause paper. The working group says it chose one broad approach to examine first, leaving other ways to implement a pause outside this initial feasibility study.

Sources: Working paper, contributors and version history ↗

AUTHORS	Working Group on AI Pause Feasibility
ORGANISATION / AFFILIATION	Multi-institution working group organized by William Fithian and Wesley H. Holliday (UC Berkeley)
ORIGINAL DOCUMENT	**The Feasibility of a Hardwired Pause of Frontier AI Training ↗**

PURPOSE

The study canvasses possible reasons for a pause, including loss of control, catastrophic misuse, gradual disempowerment, economic disruption, concentration of power, and geopolitical instability. It explicitly leaves the desirability of such a pause outside its feasibility analysis.

WHAT IS PACED

Under the studied scenario, new frontier model training stops while approved existing models can be served. New accelerator supply shifts toward inference-only chips; production of training-capable chips is halted or sharply reduced, and existing chips are inventoried.

WHEN AND FOR HOW LONG

The model explores a pause lasting at least ten years, extendable by agreement. It describes a staged transition from an initial global census and design work to replacement with inference-only hardware. Example calendar dates in the paper illustrate a scenario, not a proposed deadline.

HOW PACING WORKS

A treaty would whitelist deployable models, license and track training-capable accelerators, convert the chip supply, and limit opportunities to combine hidden compute into a frontier training cluster.

VERIFICATION METHODS

Serial-level chip registration, a global accelerator census, licensed-operator inspections, inventory reconciliation, location attestation, and checks on inference traffic and energy use are examined. Network taps, sampled recomputation, and runtime telemetry are possible additional layers.

WHO RUNS IT

The hypothetical regime needs participating states, a common authority, chip manufacturers, licensed operators, and an inspectorate with access to facilities and supply-chain records. Exact treaty governance is not the study's central design problem.

LIMITATIONS

Its feasibility depends on chip production controls, reliable inventories, and the ability to bound undeclared compute. Covert fleets, efficiency advances, repurposed hardware, and incomplete international participation remain difficult; a training pause would not by itself solve every downstream AI harm.

Mapped verification methods

10 mechanisms in the atlas map to this text. The labels indicate the strength of each match, not whether the method works.

Chip production and transfer ledger
Core

Physical chip inventory inspection
Core

Remote chip location attestation
Core

Network tap plus sampled recomputation
Specified option

Trusted execution and remote attestation
Specified option

Workload traffic fingerprinting
Specified option

Cloud workload metering
Partial match

Facility power and cooling reconciliation
Partial match

Model and checkpoint provenance registry
Partial match

Training transcript and checkpoint replay
Partial match

Explore the comparison

See all eight proposals by mechanism →

This is an analytical summary of the linked proposal. Examples and scenario dates are identified as such; see the original document for the authors' full arguments.